



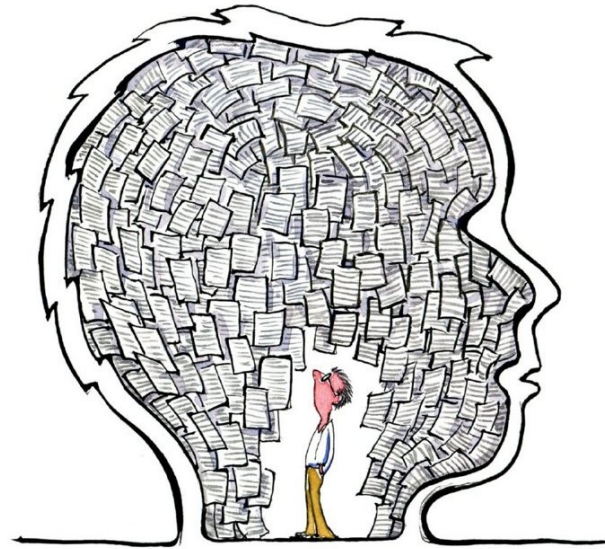
TECHNISCHE
UNIVERSITÄT
DARMSTADT

METACOGNITION: A CAPACITY THAT MAKES US SMART?

Monika Undorf

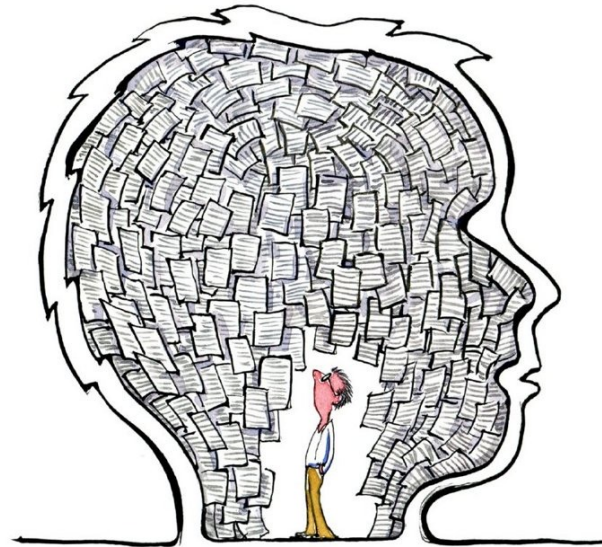
Technical University of Darmstadt

September 24, 2025



Metacognition

Knowledge about (one's own) cognitive processes and its use for regulating information processing and behavior.



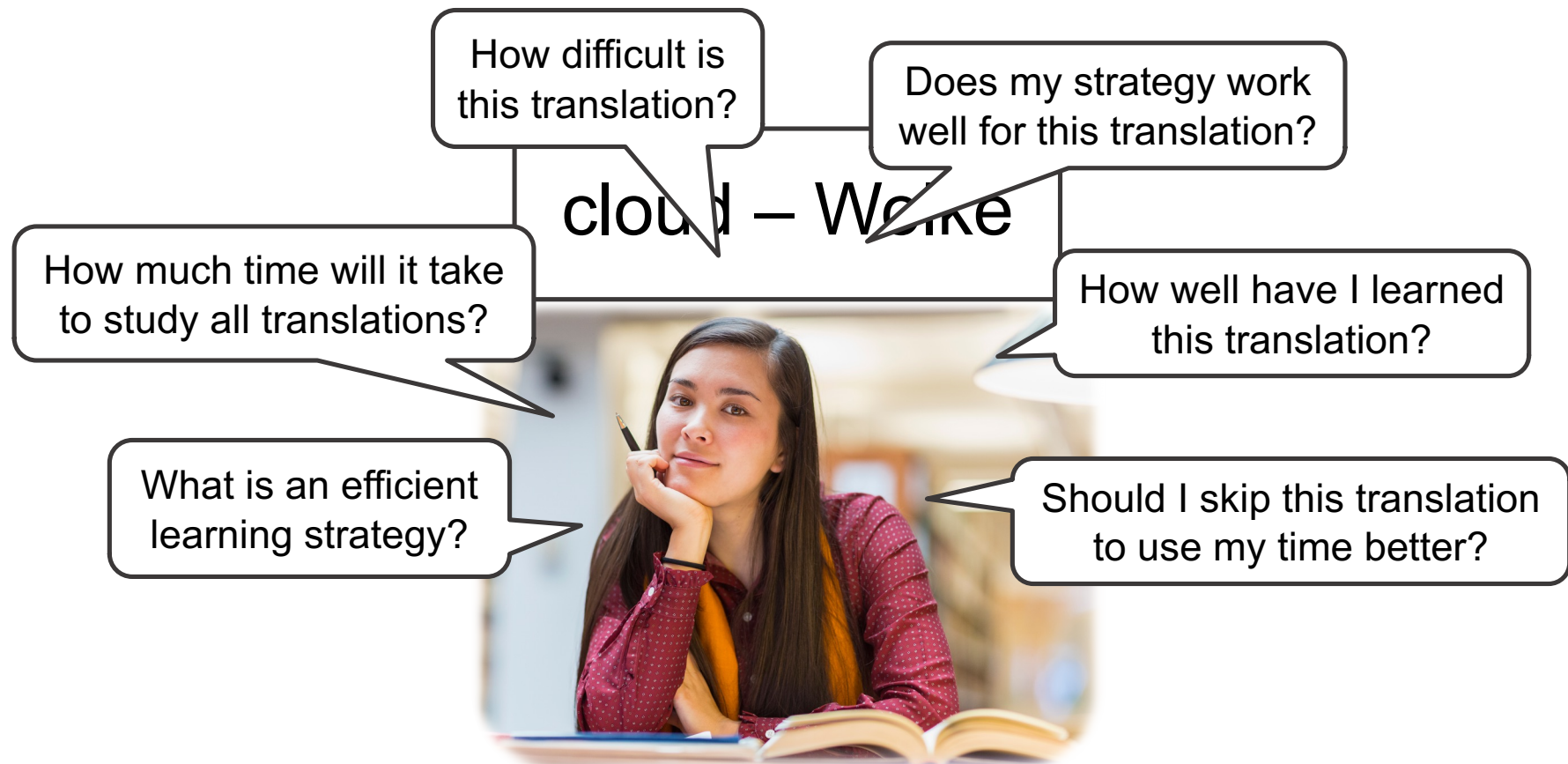
Metacognition

Monitoring

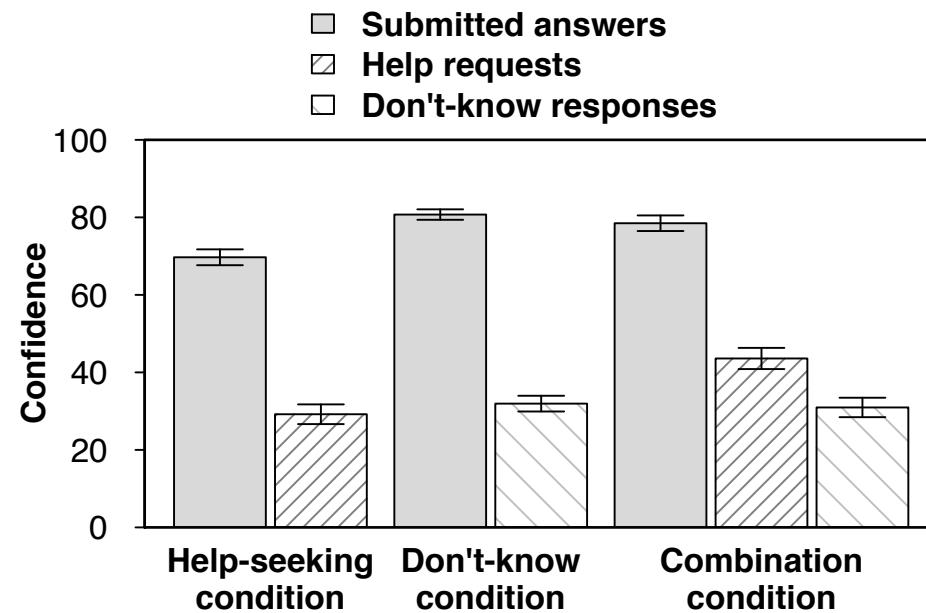
Control

Knowledge about (one's own) cognitive processes and its use for regulating information processing and behavior.

Foreign Language Learning

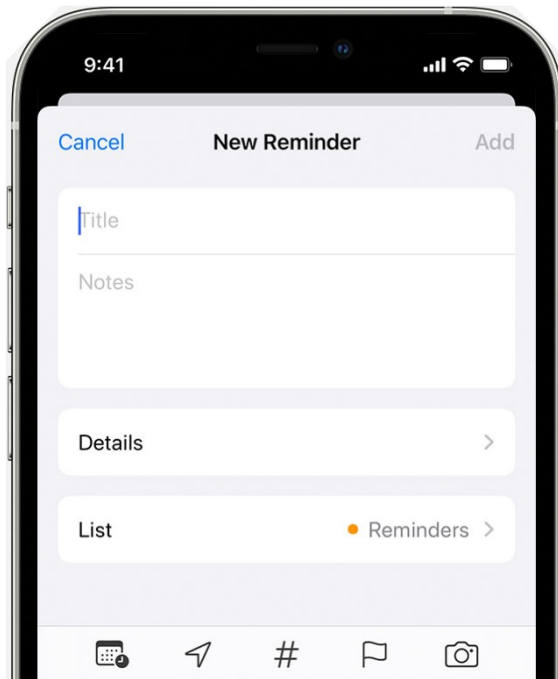


Answering Knowledge Questions



(Undorf, Livneh, & Ackerman, 2021, *Metacognition and Learning*)

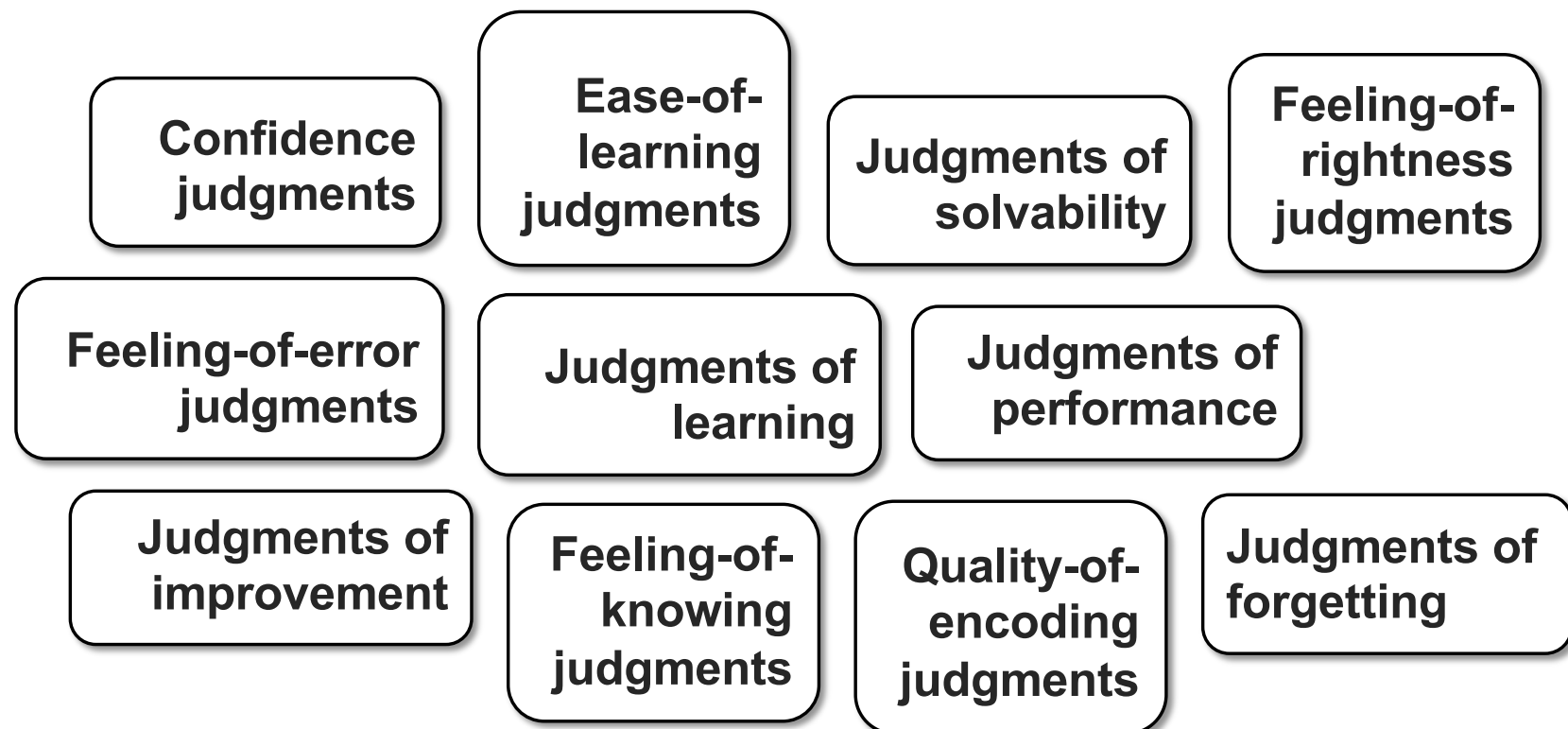
External Reminders



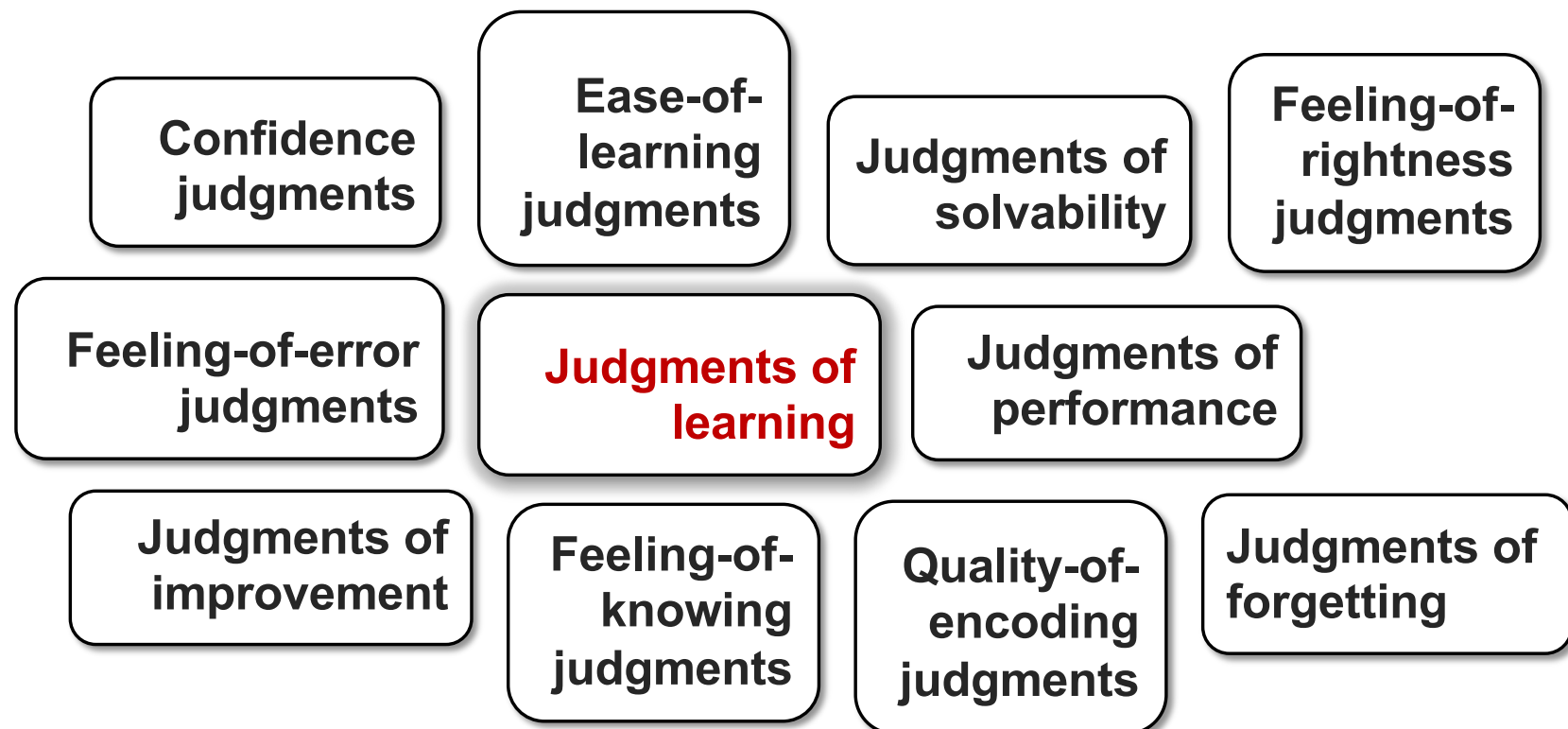
Overview

- How do people know their thoughts?
- How accurate is metacognition?
- How ubiquitous is metacognitive monitoring?
- How relevant is metacognition in real-world contexts?

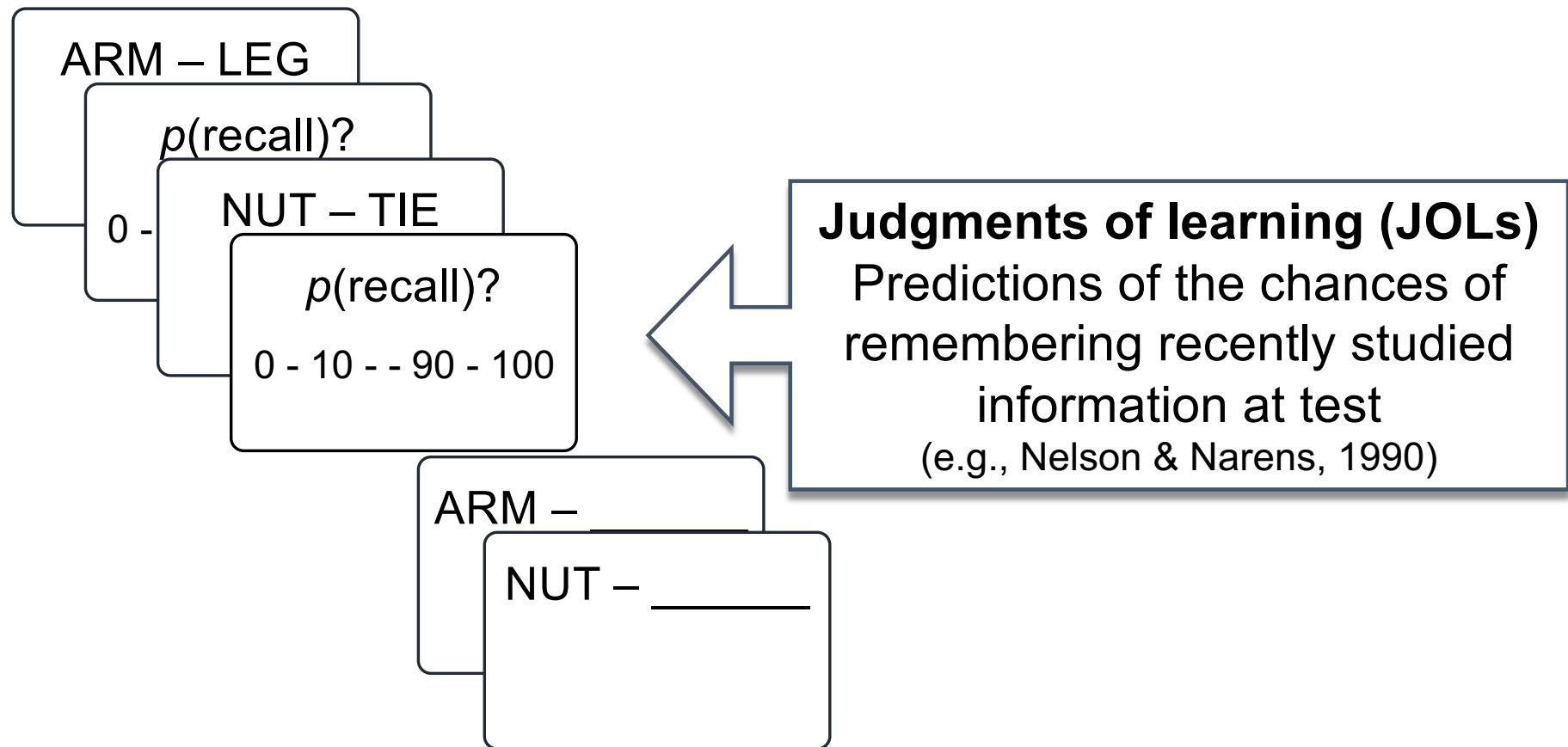
The Study of Metacognition



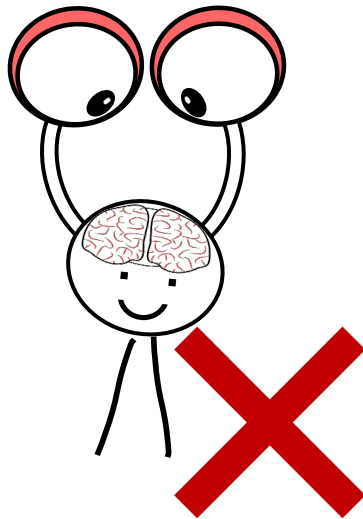
The Study of Metacognition



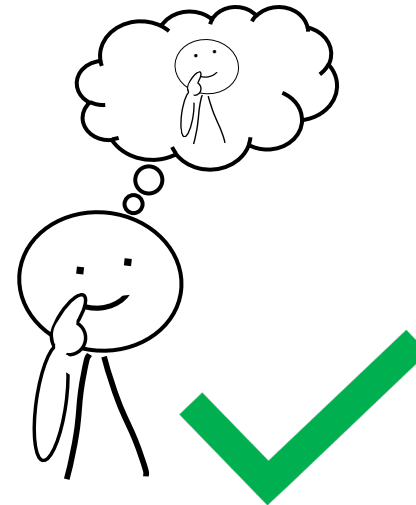
Judgments of Learning



The Basis of Metacognition



- Direct access to one's cognitions.



- Inferences based on proximal cues.

(e. g., Dunlosky & Metcalfe, 2008; Koriat, 1997; Rhodes, 2016)

Semantic Association



...

ARM — LEG

HAT — AIR

...



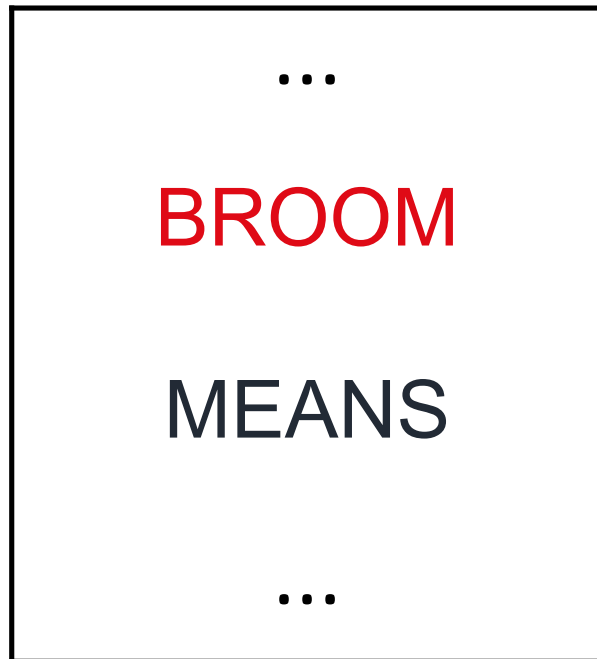
(e. g., Undorf & Erdfelder, 2015, *Memory & Cognition*)

Emotionality



(e. g., Undorf et al., 2018, *Memory & Cognition*)

Concreteness



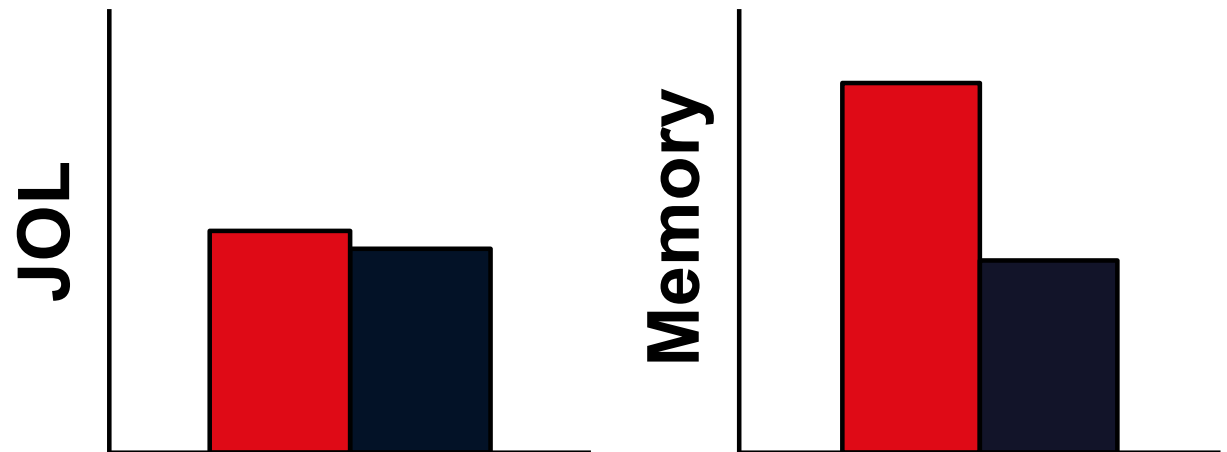
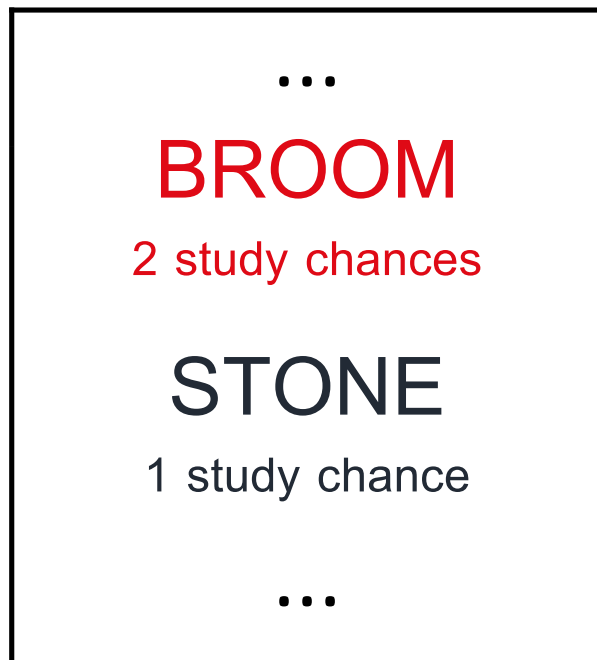
(e. g., Undorf et al., 2018, *Memory & Cognition*)

Font Size



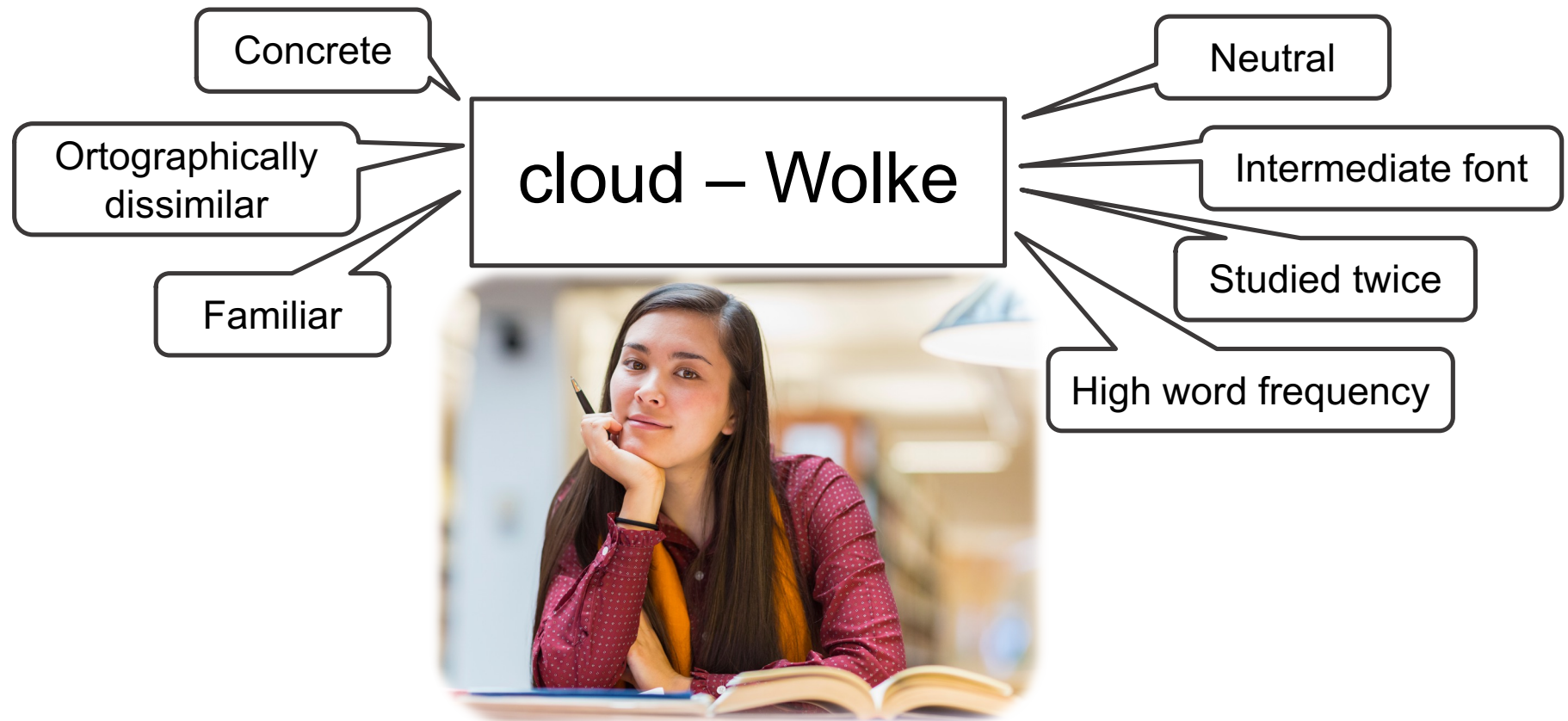
(e. g., Undorf & Zimdahl, 2018, *JEP:LMC*)

~~Future Study Opportunities~~



(e. g., Kornell & Bjork, 2009, *JEP:G*)

Multiple Cues



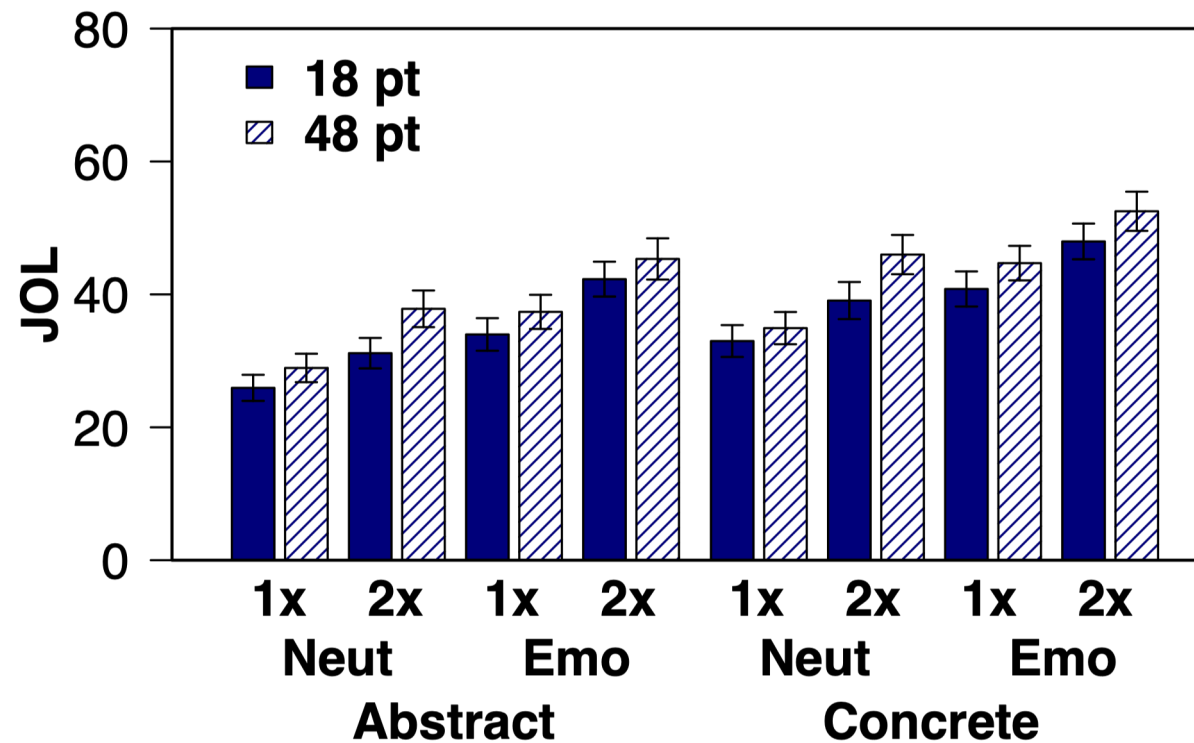
Multiple Cues



18 pt	48 pt
MEANS	RATE

(Undorf et al., 2018, *Memory & Cognition*)

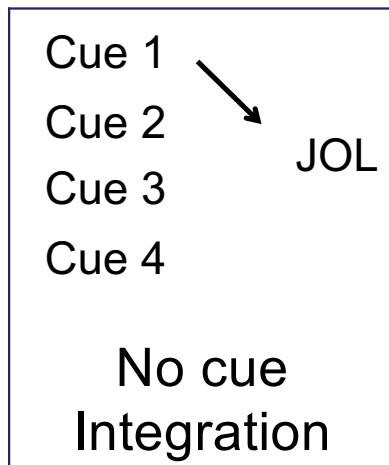
Multiple Cues



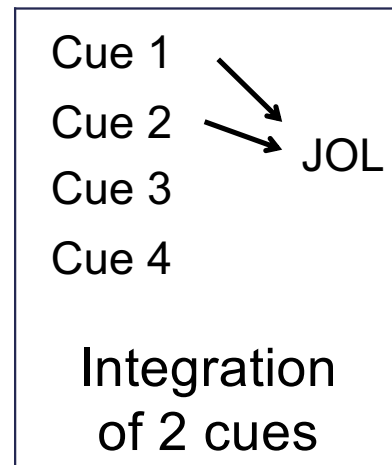
(Undorf et al., 2018, *Memory & Cognition*)

Multiple Cues

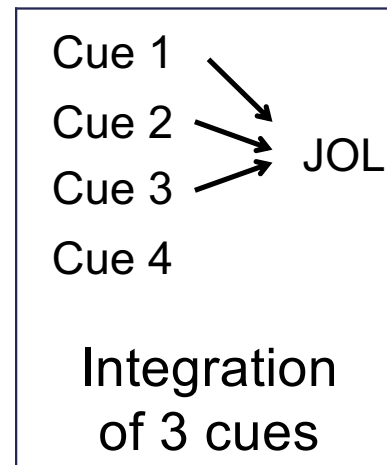
2.00%



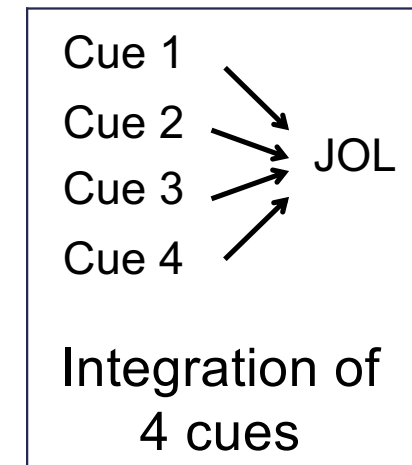
20.00%



34.00%



44.00%



(Undorf et al., 2018, *Memory & Cognition*)

Idiosyncratic Information



- **Low memorability** according to shared cues.
- **High memorability** for people who know the place.



- **Low memorability** according to shared cues.
- **High memorability** for Participant #158.

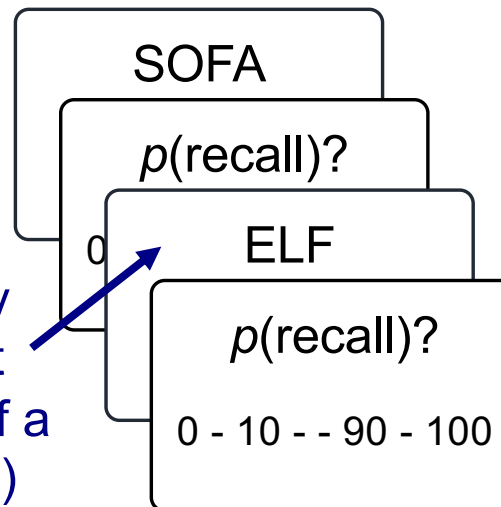
Idiosyncratic Informationen

1) Induction of personal significance

Write a story that includes:
FAIRY
SHIP
...

Personally significant
(synonym of a story word)

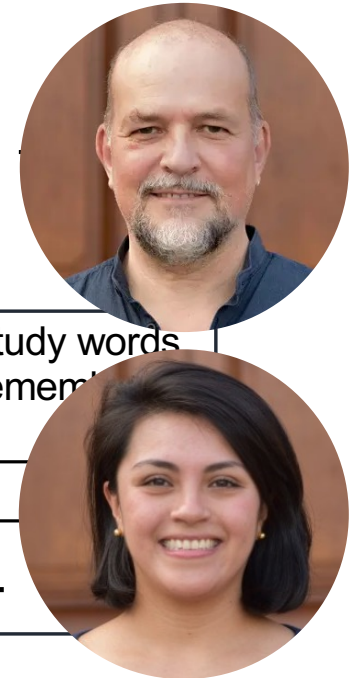
2) Study phase with JOLs



3) Free recall

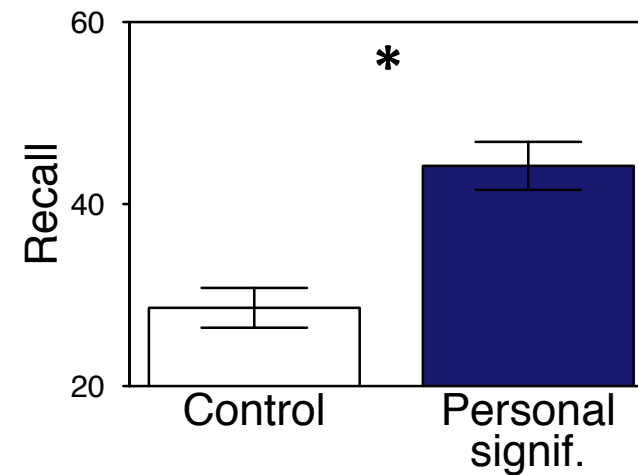
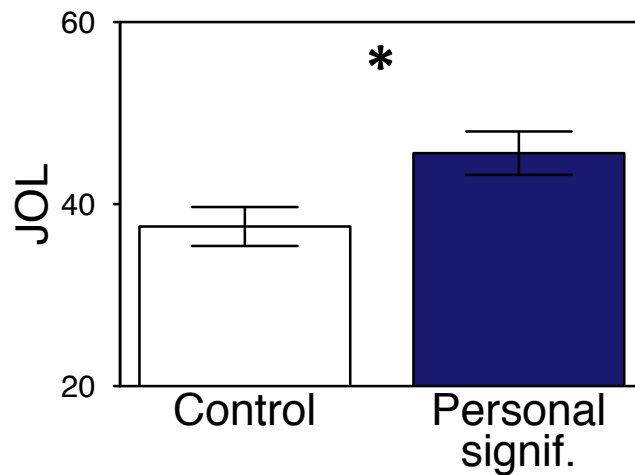
Type in all study words
you can remember

...



(Undorf, Navarro-Báez, & Bröder, 2022, *Cognition*)

Idiosyncratic Information



- Personal significance increased JOLs and memory.

(Undorf, Navarro-Báez, & Bröder, 2022, *Cognition*)

Idiosyncratic Information

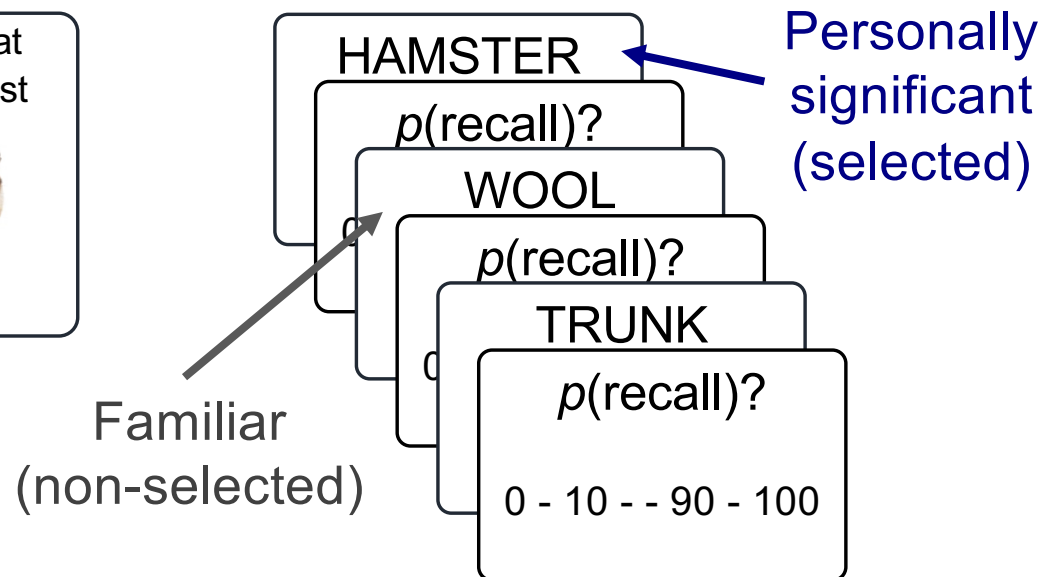
1) Assessment of personal significance

Select the picture that characterizes you best



...

2) Study phase with JOLs



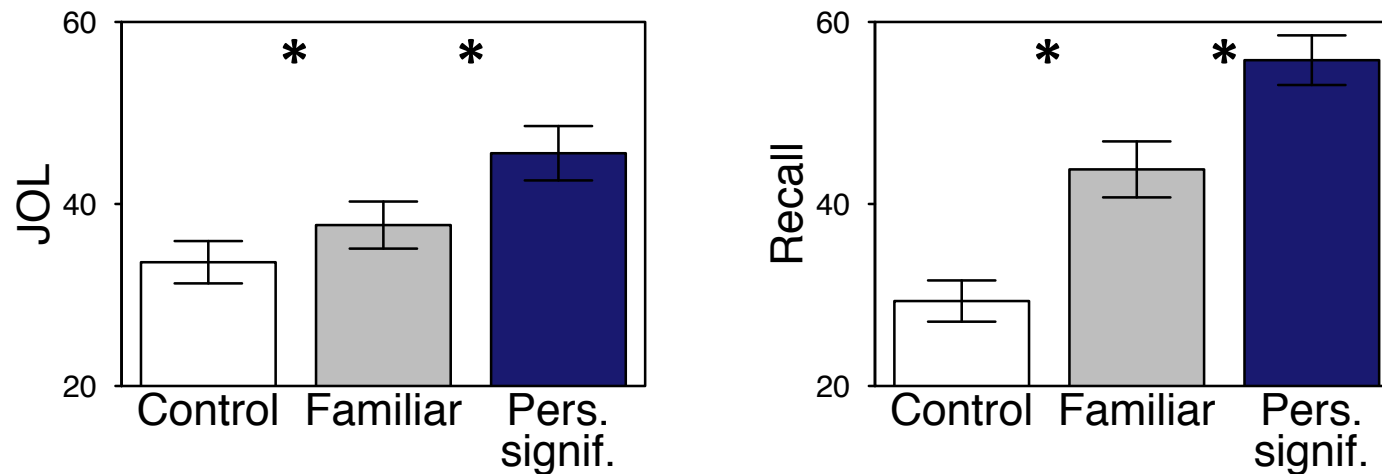
3) Free recall test

Type in all study words you can remember

...

(Undorf, Navarro-Báez, & Bröder, 2022, *Cognition*)

Idiosyncratic Information



- Personal significance increased JOLs and memory beyond familiarity.

(Undorf, Navarro-Báez, & Bröder, 2022, *Cognition*)

Overview

- How do people know their thoughts?
- How accurate is metacognition?
- How ubiquitous is metacognitive monitoring?
- How relevant is metacognition in real-world contexts?

Accuracy of Metacognition

- **Resolution:** How well do metacognitive judgments differentiate between correct vs. incorrect responses?
 - Excellent: high confidence in all correct responses, low confidence in all incorrect ones
- **Calibration:** How closely do mean metacognitive judgments match mean performance?
 - Excellent: Mean metacognitive judgments match mean performance

Accuracy of Metacognition



- Participants:
 $N = 109$ students
- 10-word summaries of text paragraphs
- Confidence judgments every 30 s and after each summary

Bitte lesen Sie den folgenden Abschnitt.

Viel Wasser und dichter Rasen bis ans Ufer: So sieht für viele Erholungssuchende das ideale Plätzchen für die heißen Sommermonate aus. Um die besten Plätze konkurrieren Erholungssuchende allerdings nicht nur mit anderen Menschen. Auch Schwäne, Enten und Gänse teilen die Vorliebe für das saftige Grün mit direktem Wasseranschluss. Mensch und Tier rücken einander auf die Pelle. Das schafft Konflikte.

Fassen Sie den Abschnitt in maximal 10 Wörtern zusammen.

0 / 10 Wörtern

Bitte lesen Sie den folgenden Abschnitt.

Mike Rutzen ist Umweltschützer und Abenteuer-Weltruhm erlangt durch seine maritimen Schutzvorrichtungen. Zu den Stars gehören Brad Pitt, Leonardo DiCaprio oder Prinz Harry. Die Biografie des Hal-Mannes befasst sich mit der Studie über Zahl und Überleben von Walen.

Fassen Sie den Abschnitt in maximal 10 Wörtern zusammen.

0 / 10 Wörtern

Bewertung Ihrer Zusammenfassung

Wie viele von 100 Punkten würden Sie für Ihre derzeitige Zusammenfassung vergeben?

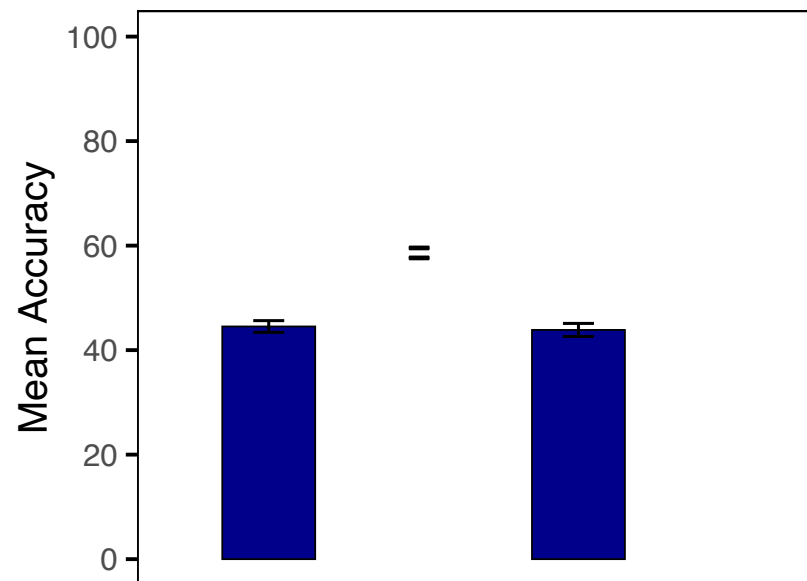
0 100

Senden

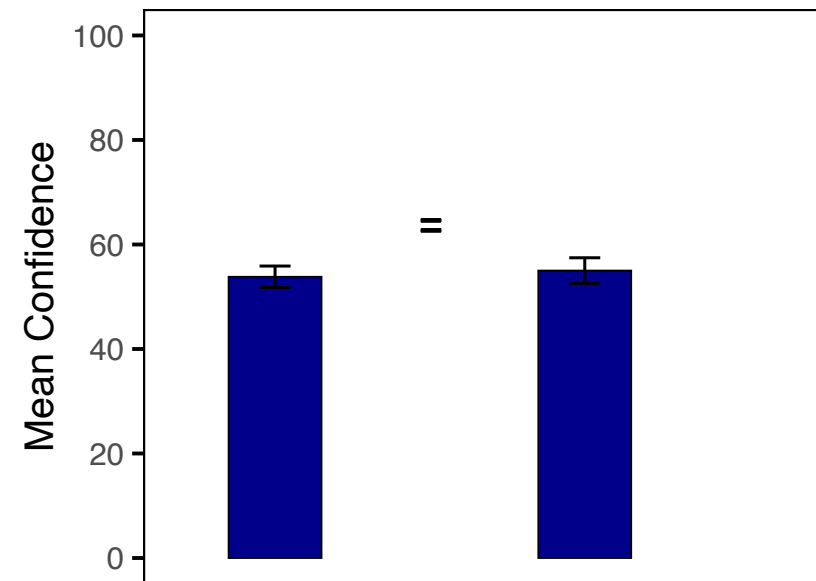
Weiter

Accuracy of Metacognition

- Actual performance:

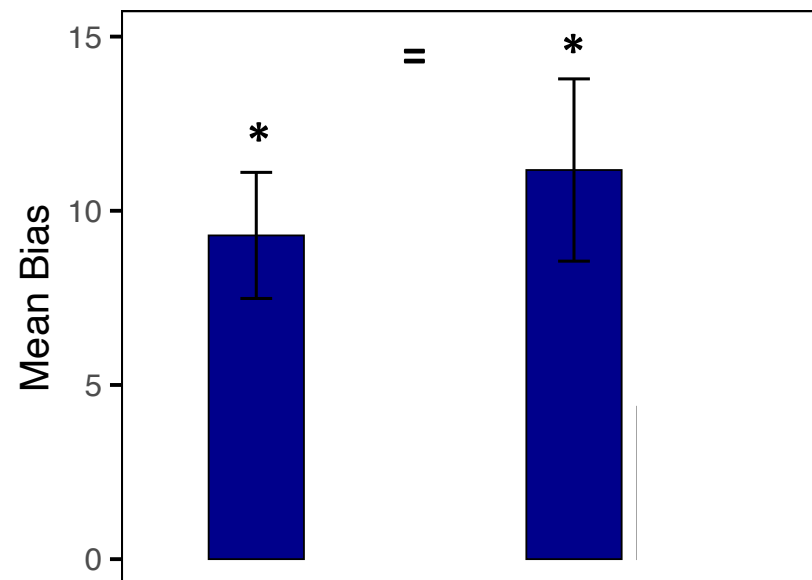


- Confidence:



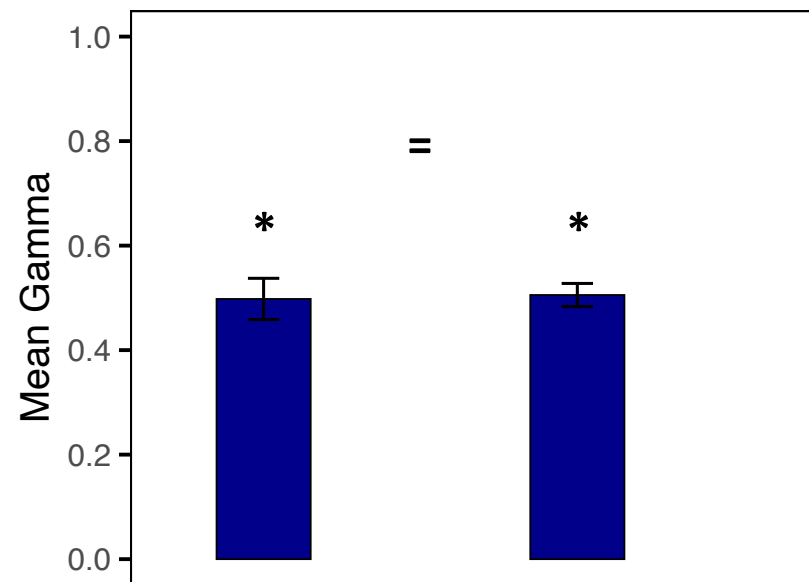
Accuracy of Metacognition

- Calibration:



Accuracy of Metacognition

- Resolution:



Accuracy of Metacognition



Human assistant

Bitte lesen Sie den folgenden Abschnitt.

Mike Rutzen ist Umweltschützer, Filmemacher und »der Hai-Mann«: Der 45-jährige Südafrikaner hat durch seine maritimen Abenteuer Weltruhm erlangt. Er ist so oft mit Weißen Haien getaucht wie kein anderer Mensch – und das ohne Schutzvorrichtungen. Zu den Gästen seiner Hai-Käfig-Touren zählen über die Jahre Prominente wie Brad Pitt, Leonardo DiCaprio oder Prinz Harry. Die italienische Zoologin Sara Andreotti von der Stellenbosch University in Südafrika war von der Biografie des Hai-Mannes beeindruckt genug, um ihn für ein ambitioniertes Forschungsprojekt anzuwerben: eine präzise Studie über Zahl und Überlebenschancen aller Weißen Haie in Südafrikas Gewässern.

Fassen Sie den Abschnitt in maximal 10 Wörtern zusammen.

0 / 10 Wörtern

Chat mit Hilfskraft

Hallo, wie kann ich Ihnen helfen?

Senden

Warten

Bewertung Ihrer Zusammenfassung

Wie viele von 100 Punkten würden Sie für Ihre derzeitige Zusammenfassung vergeben?

0 100

Senden

AI assistant

Bitte lesen Sie den folgenden Abschnitt.

Viel Wasser und dichter Rasen bis ans Ufer: So sieht für viele Erholungssuchende das ideale Plätzchen für die heißen Sommermonate aus. Um die besten Plätze konkurrieren Erholungssuchende allerdings nicht nur mit anderen Menschen. Auch Schweine, Enten und Gänse teilen die Vorliebe für das saftige Grün mit direktem Wasseranschluss. Mensch und Tier rücken einander auf die Pelle. Das schafft Konflikte.

KI-Assistent

Wie kann ich Ihnen behilflich sein?

Ihre Nachricht...

Fassen Sie den Abschnitt in maximal 10 Wörtern zusammen.

0 / 10 Wörtern

Warten

Bewertung Ihrer Zusammenfassung

Wie viele von 100 Punkten würden Sie für Ihre derzeitige Zusammenfassung vergeben?

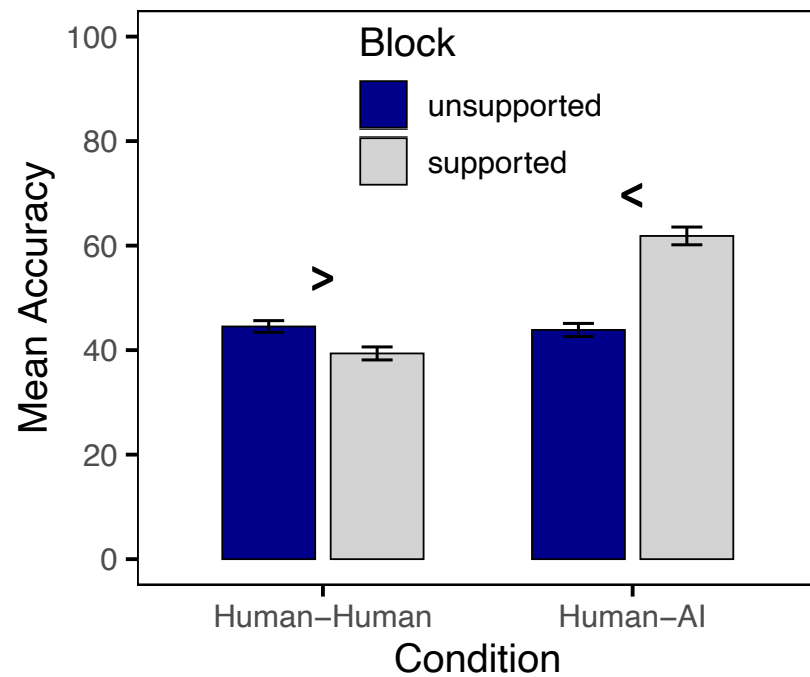
0 100

Senden

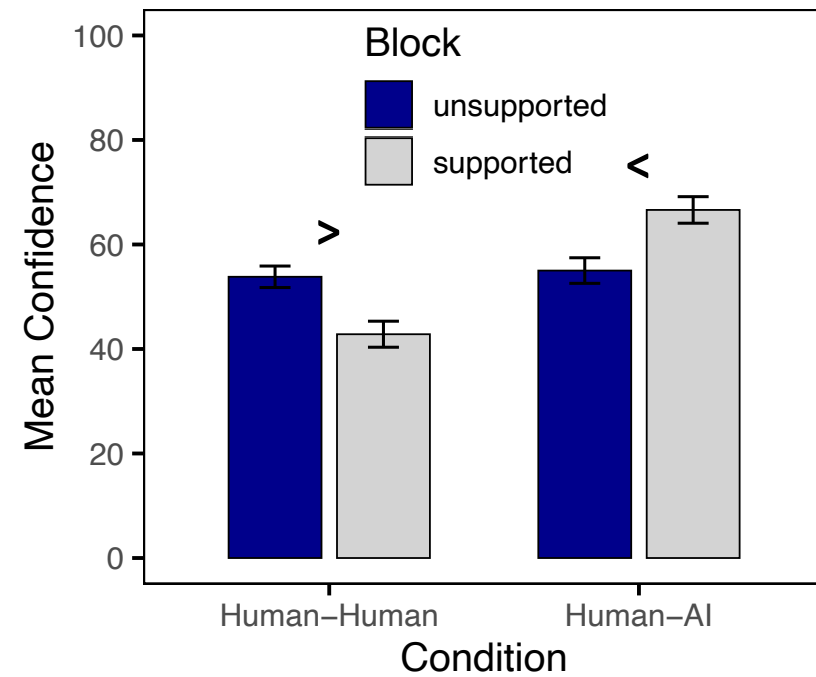
- Participants: $N = 109$ students
- 10-word summaries of text paragraphs
- Confidence judgments every 30 s and after each summary
- Block II: Supported by a human vs. AI

Accuracy of Metacognition

- Actual performance:

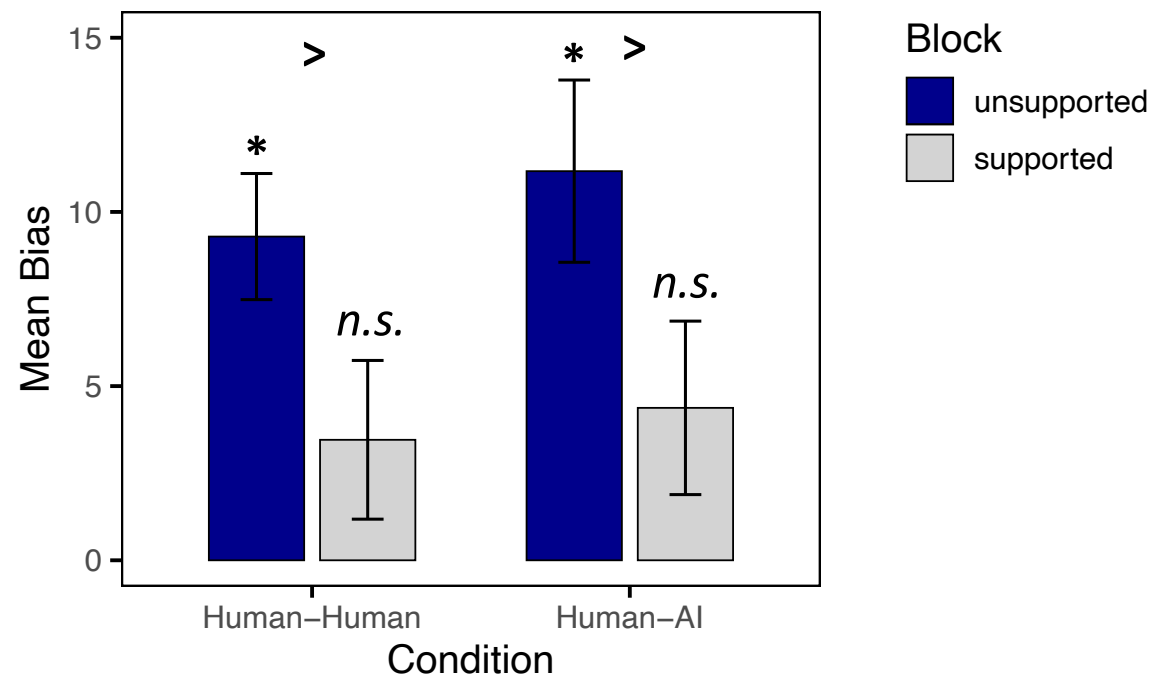


- Confidence:



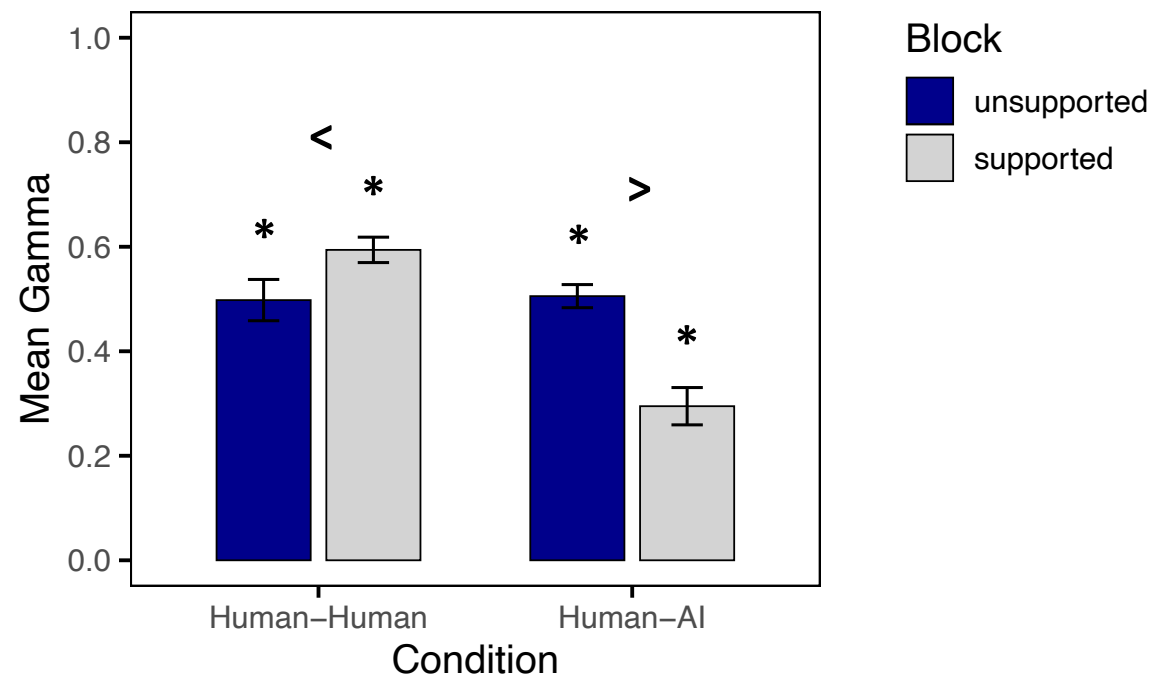
Accuracy of Metacognition

■ Calibration:



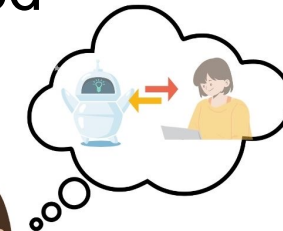
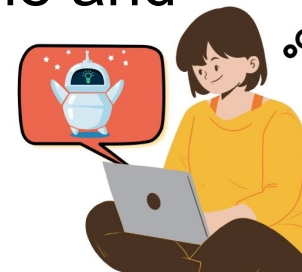
Accuracy of Metacognition

■ Resolution:



Accuracy of Metacognition

- Confidence judgments for text summaries showed above-chance resolution and good calibration.
- This held when participants worked alone and when assisted by a human or an AI.



(Undorf, Ingendahl, Rogge, Pisch, & Jussupow, in preparation)

Overview

- How do people know their thoughts?
- How accurate is metacognition?
- How ubiquitous is metacognitive monitoring?
- How relevant is metacognition in real-world contexts?



The Ubiquity Assumption

Metacognitive monitoring occurs spontaneously, constantly and inevitably during learning

JOLs only capture the ongoing monitoring processes that occur anyway

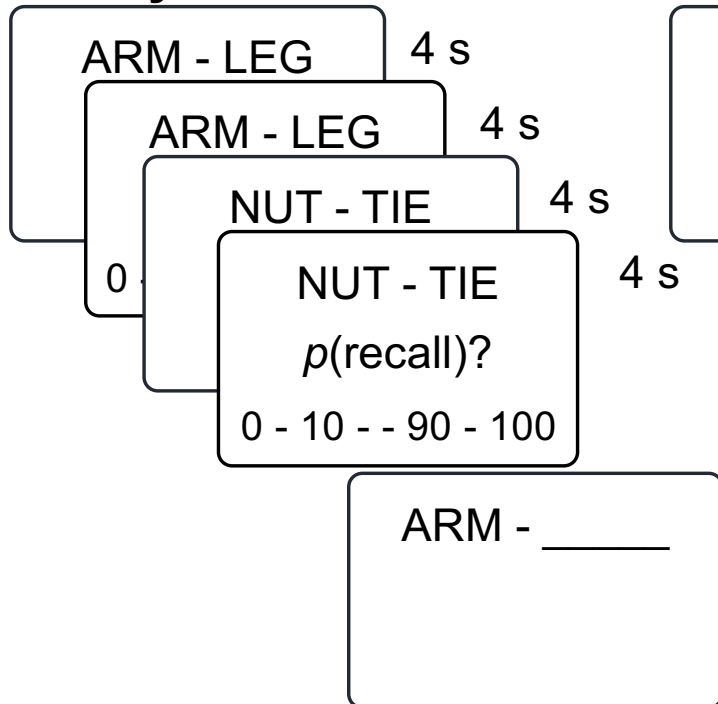


(Halamish & Undorf, 2020, *ZfP*)

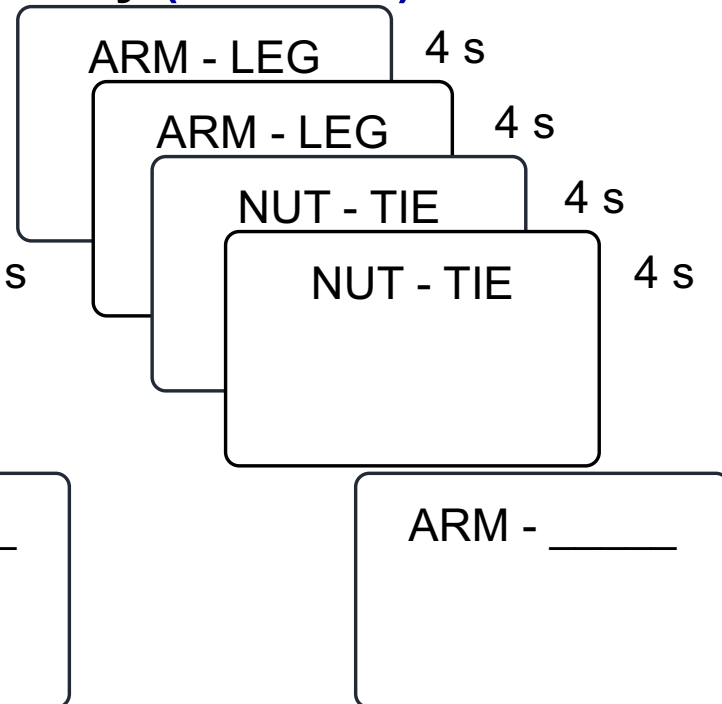
A Challenge: JOL Reactivity



Study + JOLs

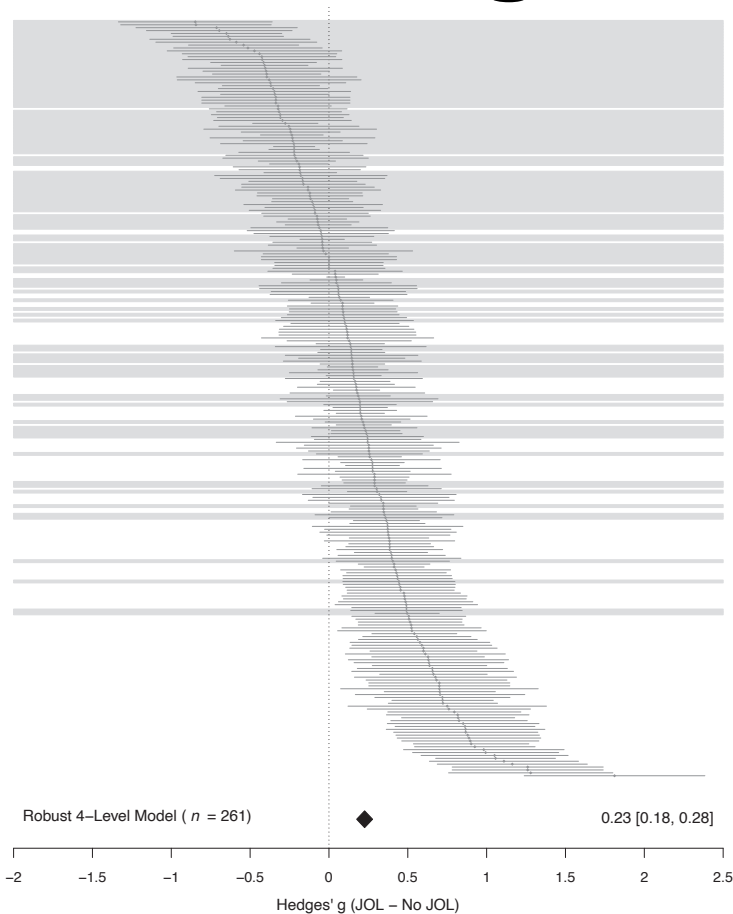


Study (no JOLs)



(Halamish & Undorf, 2020, *ZfP*)

A Challenge: JOL Reactivity



- Making JOLs ...
 - Improves cued-recall memory for related pairs (*positive reactivity*).
 - Hedge's $g = 0.44$
 - Impairs cued-recall memory for unrelated pairs (*negative reactivity*).
 - Hedge's $g = -0.09$



(Ingendahl, Halamish, & Undorf, 2025, *Psych Bull*)

Implications for Ubiquity Assumption

Monitoring **DOES NOT** occur spontaneously,
constantly and inevitably during learning

OR

Monitoring
Learning



JOL



Making JOLs **MODIFIES** the spontaneous
monitoring processes that would otherwise occur

Monitoring
Learning

JOL



Implications for Ubiquity Assumption

Monitoring **DOES NOT** occur spontaneously,
constantly and inevitably during learning

OR

Making JOLs **MODIFIES** the spontaneous
monitoring processes that would otherwise occur

Monitoring
Learning



JOL



Monitoring
Learning

JOL



Modification of Spontaneous Monitoring?

- **Overt-response hypothesis**

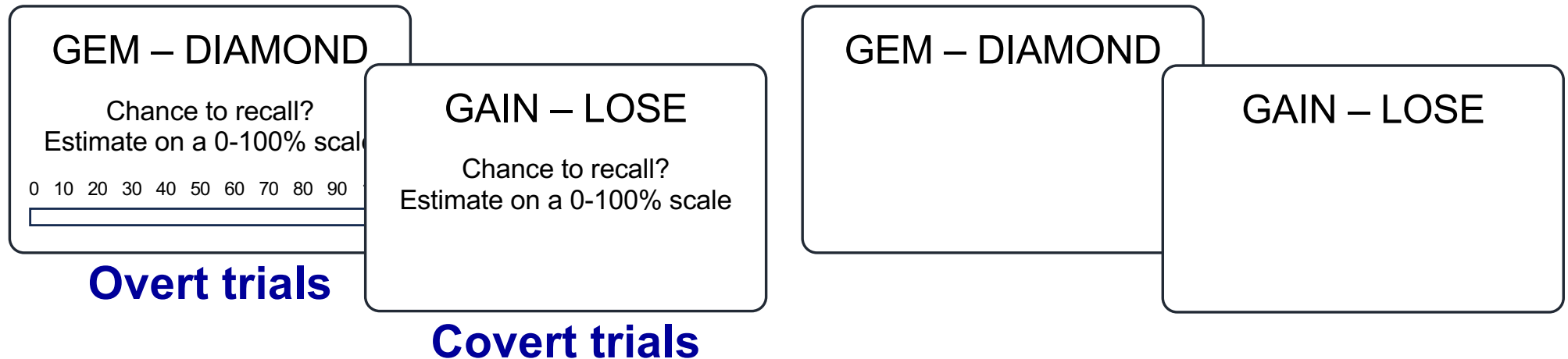
- Spontaneous monitoring is done covertly.
- Making JOLs requires overt responses.
- **Prediction:** No reactivity when making JOLs covertly.

The Role of Overt Responses

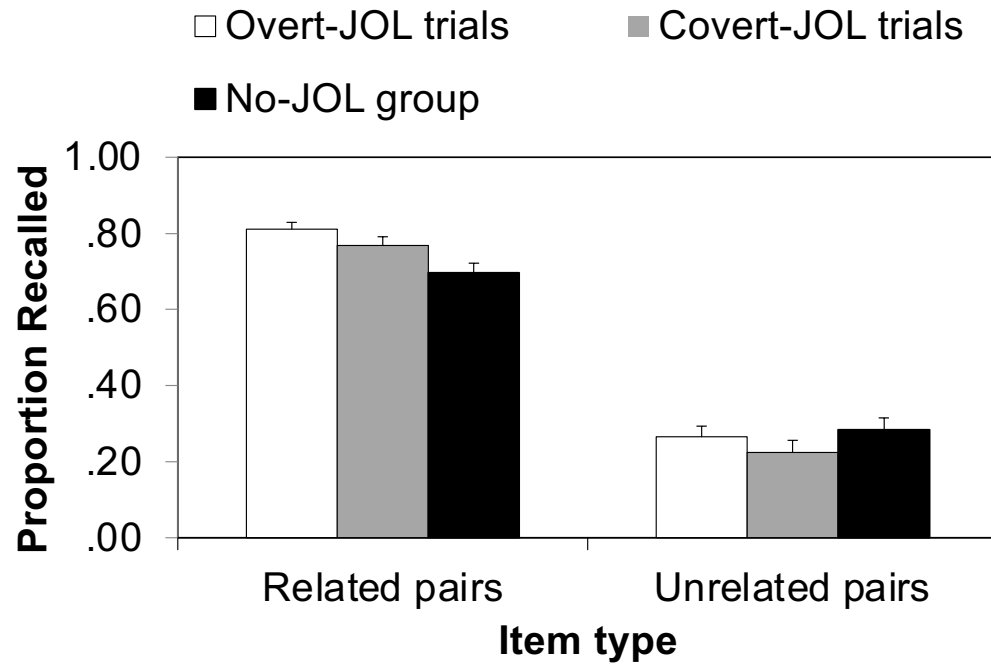
- $N = 138$ participants from Prolific
- Within-participants manipulation of response format in the JOL group.

JOL group

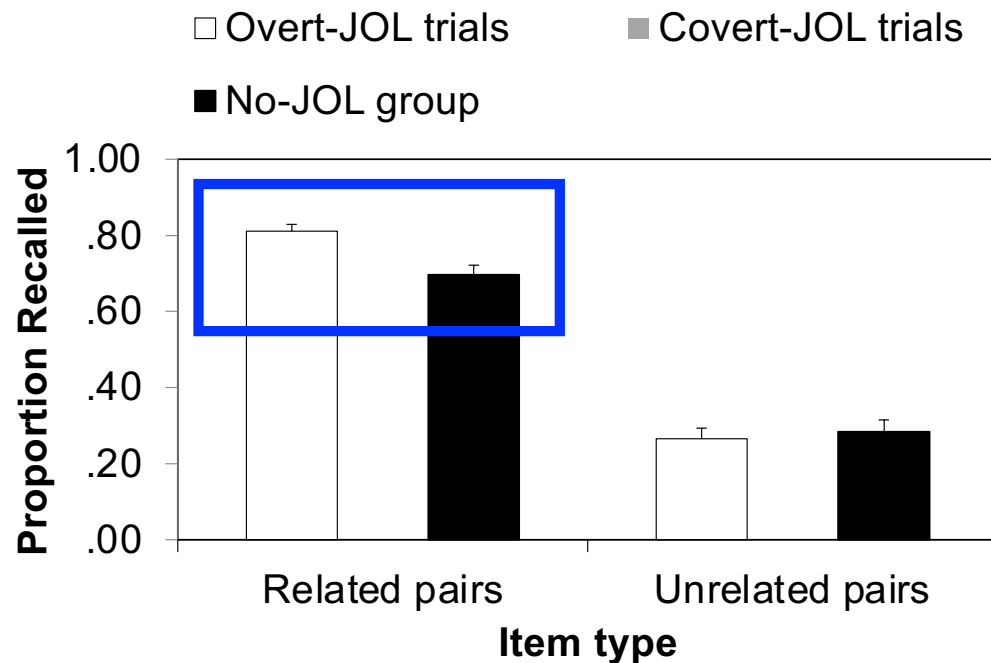
No-JOL group



The Role of Overt Responses

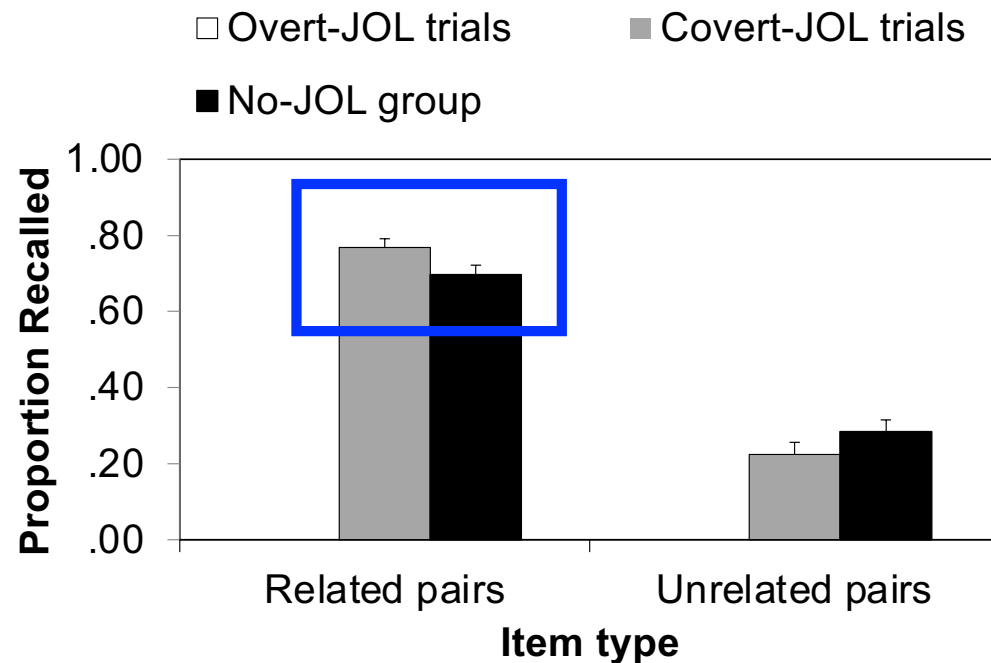


The Role of Overt Responses



- Overt-JOL trials vs. no-JOL group
 - Positive reactivity for related pairs
 - No reactivity for unrelated pairs

The Role of Overt Responses



- Overt-JOL trials vs. no-JOL group
 - Positive reactivity for related pairs
 - No reactivity for unrelated pairs
- Covert-JOL trials vs. no-JOL group
 - Positive reactivity for related pairs
 - No reactivity for unrelated pairs
- JOL reactivity is not entirely due to overt responses

Modification of Spontaneous Monitoring?

▪ Overt-response hypothesis

- Spontaneous monitoring is done covertly.
- Making JOLs requires overt responses.
- **Prediction:** No reactivity when making JOLs covertly.



Modification of Spontaneous Monitoring?

■ Overt-response hypothesis

- Spontaneous monitoring is done covertly.
- Making JOLs requires overt responses.
- **Prediction:** No reactivity when making JOLs covertly.



■ Numerical-response hypothesis

- Spontaneous monitoring is qualitative in nature.
- JOLs require a numerical response (usually %).
- **Prediction:** No reactivity when making non-numerical JOLs.

The Role of Numerical Responses

- $N = 272$
- 4 groups: 3 JOL groups with different JOL scales, 1 no-JOL group

Percentage JOLs

GEM – DIAMOND

Chance to recall?
Estimate on a 0-100% scale

0 10 20 30 40 50 60 70 80 90 100

Likert JOLs

GEM – DIAMOND

Chance to recall?

Very low Low Moderate High Very high

☐ ☐ ☐ ☐ ☐

Yes/No JOLs

GEM – DIAMOND

Will you recall?

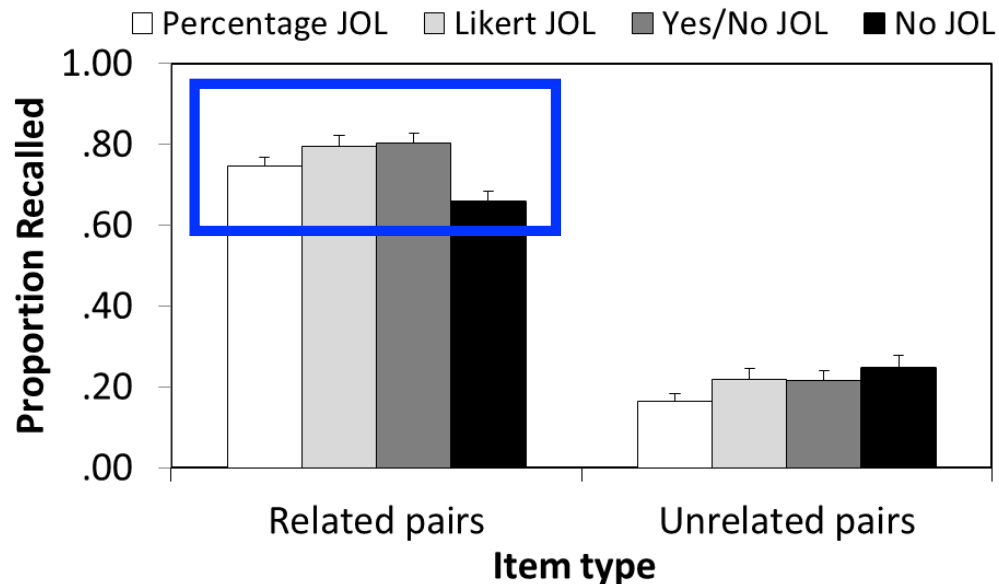
No Yes

☐ ☐

No JOLs

GEM – DIAMOND

The Role of Numerical Responses



- All JOL types yielded positive reactivity for related pairs
- No JOL type yielded reactivity for unrelated pairs
- JOL reactivity is not entirely due to numerical responses

Modification of Spontaneous Monitoring?

■ Overt-response hypothesis

- Spontaneous monitoring is done covertly.
- Making JOLs requires overt responses.
- **Prediction:** No reactivity when making JOLs covertly.



■ Numerical-response hypothesis

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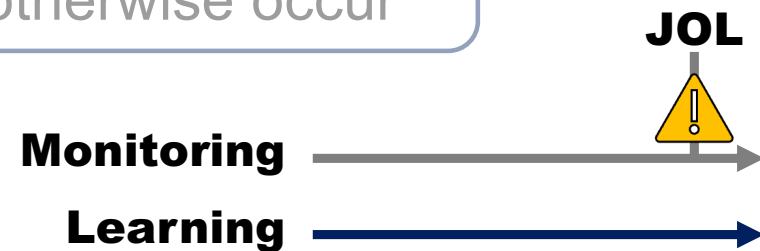
Implications for Ubiquity Assumption

Monitoring **DOES NOT** occur spontaneously,
constantly and inevitably during learning

OR



Making JOLs **MODIFIES** the spontaneous
monitoring processes that would otherwise occur



Implications for Ubiquity Assumption

- Metacognitive monitoring may not always occur spontaneously (see also Jordano & Touron, 2018).
- Spontaneous metacognitive monitoring is an important avenue for future research.



(Halamish, Meer, & Undorf, under revision)

Overview

- How do people know their thoughts?
- How accurate is metacognition?
- How ubiquitous is metacognitive monitoring?
- How relevant is metacognition in real-world contexts?

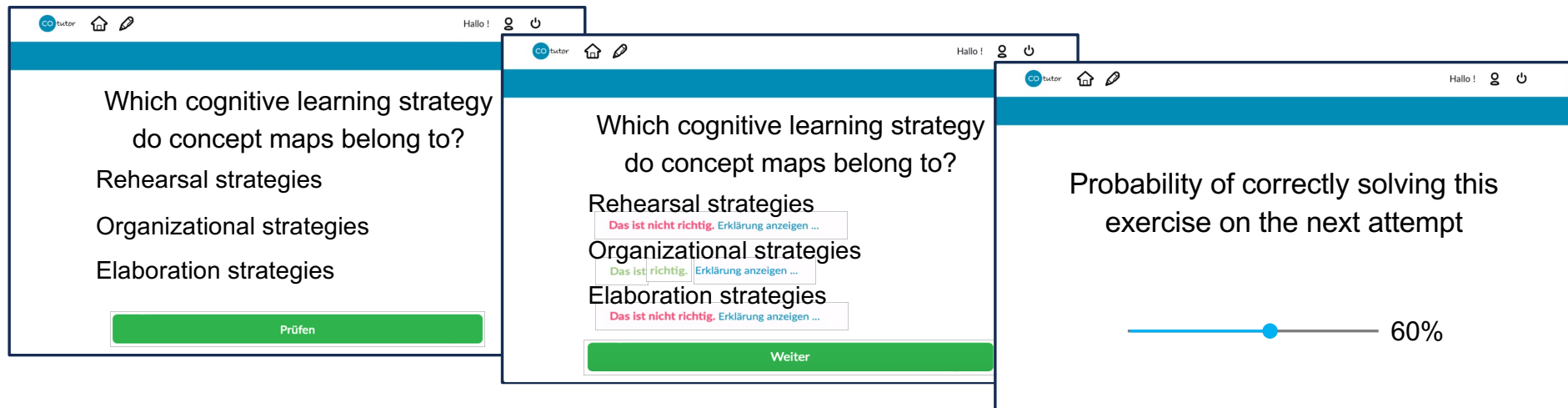
Metacognition in ITS

- Intelligent tutoring systems (ITS) ...
 - Provide study materials.
 - Analyze learners' interactions with study materials.
 - Adapt study materials to individual learners.
-  tutor: ITS for university students
 - Repeated testing as a learning strategy.
 - Variability in exercises.
 - Learners select the chapter(s) they want to study.
 - Exercises answered incorrectly are repeated more often.



Metacognition in ITS

- Participants: $N = 145$ pre-service teachers
- Lecture Educational Psychology
- JOLs for individual questions; requested for every 7th question.



The image displays three overlapping screenshots of the CO-Tutor interface, illustrating a learning exercise. Each window has a blue header bar with the CO-Tutor logo, home, and edit icons, and a status bar with 'Hallo!', a user icon, and a power icon.

Left Window: Displays the question 'Which cognitive learning strategy do concept maps belong to?' with three options: 'Rehearsal strategies', 'Organizational strategies', and 'Elaboration strategies'. A green 'Prüfen' button is at the bottom.

Middle Window: Shows the same question and options. Feedback is provided for each option: 'Rehearsal strategies' is marked 'Das ist nicht richtig. Erklärung anzeigen ...' (red), 'Organizational strategies' is marked 'Das ist richtig. Erklärung anzeigen ...' (green), and 'Elaboration strategies' is marked 'Das ist nicht richtig. Erklärung anzeigen ...' (red). A green 'Weiter' button is at the bottom.

Right Window: Displays the text 'Probability of correctly solving this exercise on the next attempt' above a progress slider. The slider is a blue line with a blue dot at the 60% mark, and the text '60%' is to the right.

Metacognition in ITS



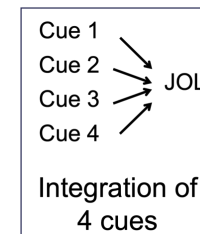
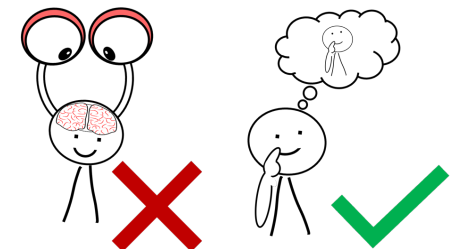
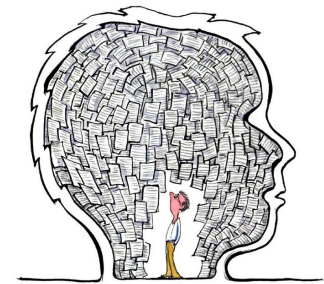
- JOLs positively predicted performance in the ITS, $OR = 3.09, p < .001$.
- Students spent more time in the ITS on medium-JOL exercises than on high- or low-JOL exercises, $\beta = -.02, p < .05$.
- High accuracy of JOLs (resolution) in the ITS predicted exam performance, $\beta = .36, p < .01$.
 - When controlling for GPA and performance in the ITS.



(Wissel, Janson, Ingendahl, Undorf, & Münzer, under revision)

Wrap-Up

- Metacognition: Knowledge about (one's own) cognitive processes and its use for self-regulation.
- People infer the state of their cognitive system from shared cues and idiosyncratic information.

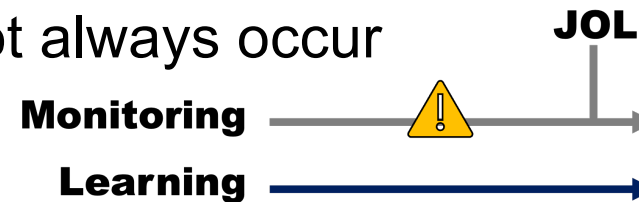


Wrap-Up

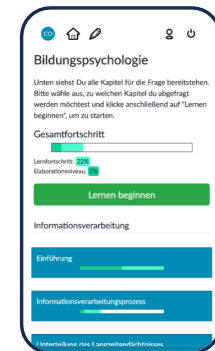
- Metacognition is moderately accurate when people work alone or collaborate with a human or an AI.



- Metacognitive monitoring might not always occur spontaneously.



- Metacognition impacts learning behavior and success when using digital learning environments.



Thank You for Your Attention!

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